

2024 Numerical Analysis Midterm

Problem 1: $f(x) = e^x - 4$

1. Prove that f has a root in $[1, 2]$, and give the bisection method algorithm.
2. Prove that after six iterations the error is less than 0.01.
3. Discuss the advantages and disadvantages of Newton's method.

Problem 2:

Consider the constrained optimization problem:

$$\min \|A - B\|_F^2 \quad \text{s.t.} \quad A\mathbf{x} = \mathbf{b}.$$

Prove that the solution exists and is unique, and that $\text{rank}(A - B) \leq 1$. Give an explicit expression for A .

Problem 3:

Given $a = x_0 < x_1 < \dots < x_n = b$ as $n + 1$ interpolation nodes,

1. Write the Lagrange interpolation basis function $L_i(x)$ (i.e., $L_i(x) \in P_n$, and $L_i(x_j) = \delta_{ij}$).
2. Prove that $\sum_{j=0}^n L_j(x) = 1$.
3. Using the above identity, derive the barycentric formula for Lagrange interpolation. Give the computational complexity of the barycentric formula.

Problem 4:

Given nodes $0 = x_0 < x_1 < \dots < x_n = 1$, consider the spline function on the node set $\{x_i\}_{i=0}^n$. If $S(x)$ is the natural cubic spline interpolant of the function $\phi(x) \in C^2[0, 1]$, prove the energy norm inequality:

$$\int_0^1 |\phi''(x)|^2 dx \geq \int_0^1 |S''(x)|^2 dx.$$

Problem 5:

Given points $X = \{x_i\}_{i=0}^n \subseteq [a, b]$, where the x_i are distinct. Prove that the following two definitions of the Lebesgue constant are equivalent:

1. $\Lambda_n(X) = \max_{x \in [a, b]} \sum_{i=0}^n |L_i(x)|$, where $L_i(x)$ are the Lagrange basis functions.
2. $\Lambda_n(X) = \sup \{ \max_{x \in [a, b]} |P_n(x)| : P_n(x) \in \mathcal{P}_n, \max |y_i| \leq 1 \}$, where $P_n(x)$ satisfies the interpolation conditions $P_n(x_i) = y_i$, $i = 0, 1, \dots, n$.

Problem 6:

If p_k^* is the best uniform approximation of f on $[-1, 1]$, and f is an even function on $[-1, 1]$,

1. Prove that p_k^* is even and

$$p_{2m}^* = p_{2m+1}^*$$

2. Prove or disprove that 0 is an extremal point of $f - p_k^*$.

Problem 7:

Discuss the drawbacks of high-degree Lagrange interpolation with equally spaced nodes, and propose solutions.